Nonlinear Finite Element Methods PVL

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1 Introduction

This report presents the finite element implementation of an axisymmetric thick-walled pipe subjected to internal pressure. Due to the rotational symmetry of the problem, the deformation is described only by the radial displacement field $u_r(r)$. The corresponding stress state is represented by the radial stress σ_{rr} and the hoop stress $\sigma_{\varphi\varphi}$.

The material is modelled as elastic ideal-plastic with the Tresca yield criterion. The finite element formulation is based on linear one-dimensional elements in the radial direction. The nonlinear elastic-plastic problem is solved incrementally by applying the internal pressure from zero up to the maximum pressure $p_{\max}$. For each load increment, the equilibrium equations are solved using the Newton-Raphson method. The plastic strain is treated as an internal state variable and the plastic strain at the Gauss points are stored.

The implementation is first verified in the purely elastic regime by comparing the numerical solution with the analytical displacement solution for a thick-walled pipe. Subsequently, the full elastic-plastic response is investigated at the maximum pressure. Mesh convergence and load-increment convergence studies are performed, and the final displacement field, stress distribution, Tresca stress distribution, pipe widening history and plastic zone development are evaluated.

2 Problem Definition

The considered boundary value problem is a thick-walled pipe with inner radius a and outer radius b, subjected to internal pressure p. The outer boundary is traction-free. Therefore, the radial stress boundary conditions are

$$\sigma_{rr}(r = a) = -p, \qquad \sigma_{rr}(r = b) = 0.$$

The implemented variant is variant 4. The corresponding material, geometry and loading parameters are listed in Table 1.

Table 1: Material, geometry and loading parameters for variant 4.

Parameter	Value
Plane condition	Plane stress (PS)
Number of Gauss points	1
Young's modulus E	200 000 MPa
Poisson's ratio ν	0.30
Yield stress σ_y	200 MPa
Inner radius a	30 mm
Outer radius b	60 mm
Maximum pressure $p_{\max}$	120 MPa

A locally refined radial mesh is used, with smaller elements close to the inner radius. The mesh refinement factor is set to 2, so that the element size near the outer radius is twice the element size near the inner radius.

The pressure at which plasticity is initiated is given by

$$p_{\text{init}} = \sigma_y \frac{b^2 - a^2}{2b^2}.$$

For the present variant, this gives

$$p_{\text{init}} = 200 \frac{60^2 - 30^2}{2 \cdot 60^2} = 75\,\text{MPa}.$$

For the elastic verification, a pressure below the plastic initiation pressure is used:

$$p_{\text{elastic}} = 0.5 p_{\text{init}} = 37.5\,\text{MPa}.$$

The full elastic-plastic simulation is performed by increasing the pressure incrementally from zero to

$$p_{\max} = 120\,\text{MPa}.$$

Since $p_{\max} > p_{\text{init}}$, plastic deformation is expected during the final loading stage.

3 Governing Equations and Spatial Discretization

3.1 Weak form

For the axisymmetric thick-walled pipe problem, the only non-vanishing equilibrium equation is

$$0 = \frac{\partial (r\sigma_{rr})}{\partial r} - \sigma_{\varphi\varphi}.$$

The weak form is obtained by multiplying the equilibrium equation with the virtual radial displacement δu_r and integrating over the radial domain. This gives

$$\delta W = \int_a^b \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma}\, r\, dr - [r\sigma_{rr}\delta u_r]_{r=a}^{r=b} = 0.$$

The stress vector is written as

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{rr} \\ \sigma_{\varphi\varphi} \end{bmatrix},$$

and the strain vector is

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{rr} \\ \varepsilon_{\varphi\varphi} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial u_r}{\partial r} \\ \dfrac{u_r}{r} \end{bmatrix}.$$

The factor r in the integral comes from the axisymmetric geometry. It accounts for the increasing circumference at larger radius.

3.2 Spatial discretization

The radial displacement inside one finite element is approximated using linear shape functions:

$$u_r^e(r) = \mathbf{N}^T \hat{\mathbf{u}}^e.$$

The element nodal displacement vector is

$$\hat{\mathbf{u}}^e = \begin{bmatrix} u_1^e \\ u_2^e \end{bmatrix}.$$

The linear shape functions in the parent domain $\xi \in [-1, 1]$ are

$$\mathbf{N} = \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(1-\xi) \\ \frac{1}{2}(1+\xi) \end{bmatrix}.$$

The physical radial coordinate is interpolated in the same way:

$$r(\xi) = \mathbf{N}^T \mathbf{r}^e,$$

where

$$\mathbf{r}^e = \begin{bmatrix} r_1^e \\ r_2^e \end{bmatrix}.$$

The Jacobian of the mapping is

$$J = \frac{dr}{d\xi}.$$

Using the chain rule, the derivatives of the shape functions with respect to the physical coordinate are computed by

$$\frac{\partial N_i}{\partial r} = \frac{\partial N_i}{\partial \xi} \frac{1}{J}.$$

The strain-displacement relation is written as

$$\boldsymbol{\varepsilon} = \mathbf{B} \hat{\mathbf{u}}^e,$$

with the element B-matrix

$$\mathbf{B} = \begin{bmatrix} \frac{\partial N_1}{\partial r} & \frac{\partial N_2}{\partial r} \\ \frac{N_1}{r} & \frac{N_2}{r} \end{bmatrix}.$$

4 Material Model and Return Mapping

4.1 Elastic constitutive law

For the implemented variant, plane stress conditions are considered. Therefore, the elastic material stiffness matrix is

$$\mathbf{C} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix}.$$

The elastic stress-strain relation is

$$\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}.$$

For the elastic-plastic material model, the total strain is split into an elastic part and a plastic part:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^{pl}.$$

Thus, the elastic strain is

$$\boldsymbol{\varepsilon}^e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{pl}.$$

The plastic strain is treated as an internal state variable. It is stored at the Gauss points and updated after convergence of each load increment.

4.2 Tresca yield function

For the considered axisymmetric stress state, the Tresca equivalent stress is

$$\sigma_{\mathrm{eq}} = \left|\sigma_{rr} - \sigma_{\varphi\varphi}\right|.$$

The yield function is

$$\Phi = \sigma_{\mathrm{eq}} - \sigma_y \leq 0.$$

If $\Phi < 0$, the material response is elastic. If $\Phi = 0$, the stress state lies on the yield surface.

4.3 Elastic predictor

At the beginning of a load increment, the plastic strain from the previous converged increment is known as $\boldsymbol{\varepsilon}_n^{pl}$. For a given total strain $\boldsymbol{\varepsilon}_{n+1}$, the trial elastic strain is

$$\boldsymbol{\varepsilon}^{e,\mathrm{tr}} = \boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\varepsilon}_n^{pl}.$$

The trial stress is computed as

$$\boldsymbol{\sigma}^{\mathrm{tr}} = \mathbf{C}\boldsymbol{\varepsilon}^{e,\mathrm{tr}}.$$

The trial yield function is

$$\Phi^{\mathrm{tr}} = \left|\sigma_{rr}^{\mathrm{tr}} - \sigma_{\varphi\varphi}^{\mathrm{tr}}\right| - \sigma_y.$$

If

$$\Phi^{\mathrm{tr}} \leq 0,$$

the trial state is admissible and the step is elastic. In this case,

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}^{\mathrm{tr}}, \qquad \boldsymbol{\varepsilon}_{n+1}^{pl} = \boldsymbol{\varepsilon}_n^{pl}, \qquad \mathbf{C}_t = \mathbf{C}.$$

4.4 Plastic corrector

If

$$\Phi^{\text{tr}} > 0,$$

the trial stress violates the yield condition and a plastic correction is required. The flow direction is

$$\mathbf{n} = \text{sign}\left(\sigma_{rr}^{\text{tr}} - \sigma_{\varphi\varphi}^{\text{tr}}\right) \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Using the backward Euler update, the plastic strain is updated as

$$\boldsymbol{\varepsilon}_{n+1}^{pl} = \boldsymbol{\varepsilon}_{n}^{pl} + \Delta\lambda \mathbf{n},$$

where $\Delta\lambda$ is the plastic multiplier increment.
The corrected stress is

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}^{\text{tr}} - \Delta\lambda \mathbf{C}\mathbf{n}.$$

The corrected stress has to satisfy the consistency condition

$$\Phi_{n+1} = 0.$$

This gives the plastic multiplier increment

$$\Delta\lambda = \frac{\Phi^{\text{tr}}}{\mathbf{n}^T \mathbf{C}\mathbf{n}}.$$

Therefore, the stress update in the plastic case is

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}^{\text{tr}} - \frac{\Phi^{\text{tr}}}{\mathbf{n}^T \mathbf{C}\mathbf{n}} \mathbf{C}\mathbf{n}.$$

4.5 Algorithmically consistent tangent

For the Newton-Raphson method, the material tangent must be consistent with the stress update. For the plastic case, the algorithmically consistent tangent is

$$\mathbf{C}_t = \mathbf{C} - \frac{(\mathbf{C}\mathbf{n})(\mathbf{n}^T \mathbf{C})}{\mathbf{n}^T \mathbf{C}\mathbf{n}}.$$

For an elastic material step, the tangent remains equal to the elastic stiffness matrix:

$$\mathbf{C}_t = \mathbf{C}.$$

5 Element and Global System

5.1 Element internal force and tangent stiffness

Using the finite element discretization, the discrete weak form can be written as

$$\delta W = \delta \hat{\mathbf{u}}^T \left[\hat{\mathbf{F}}_{\text{int}} - \hat{\mathbf{F}}_{\text{ext}}\right] = 0.$$

Therefore, the global residual vector is

$$\hat{\mathbf{R}} = \hat{\mathbf{F}}_{\text{int}} - \hat{\mathbf{F}}_{\text{ext}}.$$

For one finite element, the internal force vector is

$$\hat{\mathbf{F}}^e_{\text{int}} = \int_{\Omega^e} \mathbf{B}^T \boldsymbol{\sigma}\, r\, dr.$$

After transforming the integral from the physical coordinate r to the parent coordinate ξ, this becomes

$$\hat{\mathbf{F}}^e_{\text{int}} = \int_{-1}^{1} \mathbf{B}^T \boldsymbol{\sigma}\, r J\, d\xi.$$

For the present variant, one Gauss point is used. Hence,

$$\hat{\mathbf{F}}^e_{\text{int}} \approx \mathbf{B}^T \boldsymbol{\sigma}\, r J w, \qquad \xi = 0, \qquad w = 2.$$

The element tangent stiffness matrix is obtained by linearizing the internal force vector with respect to the element nodal displacement vector:

$$\mathbf{K}^e_t = \frac{\partial \hat{\mathbf{F}}^e_{\text{int}}}{\partial \hat{\mathbf{u}}^e}.$$

Using

$$d\boldsymbol{\sigma} = \mathbf{C}_t d\boldsymbol{\varepsilon} = \mathbf{C}_t \mathbf{B} d\hat{\mathbf{u}}^e,$$

the element tangent stiffness matrix is

$$\mathbf{K}^e_t = \int_{-1}^{1} \mathbf{B}^T \mathbf{C}_t \mathbf{B}\, r J\, d\xi.$$

With one Gauss point, this is evaluated as

$$\mathbf{K}^e_t \approx \mathbf{B}^T \mathbf{C}_t \mathbf{B}\, r J w.$$

5.2 External force vector

The pipe is loaded only at the inner boundary by the internal pressure p. The outer boundary is traction-free. Therefore, the external nodal force vector contains only the contribution at the first node:

$$\hat{\mathbf{F}}_{\text{ext}} = \begin{bmatrix} ap & 0 & \cdots & 0 \end{bmatrix}^T.$$

5.3 Assembly and internal state variables

The global internal force vector and global tangent stiffness matrix are assembled from the element contributions. For element e, the local element displacement vector is extracted from the global displacement vector using the assignment matrix $\mathbf{A}^e$:

$$\hat{\mathbf{u}}^e = \mathbf{A}^e \hat{\mathbf{u}}.$$

The global internal force vector is assembled by

$$\hat{\mathbf{F}}_{\text{int}} = \sum_e (\mathbf{A}^e)^T \hat{\mathbf{F}}^e_{\text{int}}.$$

The global tangent stiffness matrix is assembled by

$$\mathbf{K}_t = \sum_e (\mathbf{A}^e)^T \mathbf{K}_t^e \mathbf{A}^e.$$

The nodal displacement vector is stored at the nodes. In contrast, stresses, plastic strains and plastic flags are evaluated and stored at the Gauss points. For the present variant, one Gauss point is used per element. Therefore, for n_{elem} elements, the plastic strain history is stored as a Gauss-point quantity:

$$\boldsymbol{\varepsilon}^{pl} = \boldsymbol{\varepsilon}^{pl}(e, gp), \qquad e = 1, \ldots, n_{\text{elem}}, \qquad gp = 1.$$

At each Gauss point, two plastic strain components are stored:

$$\boldsymbol{\varepsilon}^{pl} = \begin{bmatrix} \varepsilon_{rr}^{pl} \\ \varepsilon_{\varphi\varphi}^{pl} \end{bmatrix}.$$

After convergence of a load increment, the updated plastic strain field is used as the old plastic strain field for the next load increment. This transfer of internal state variables is required because the elastic-plastic response is path-dependent.

6 Nonlinear Solution Procedure

The elastic-plastic problem is solved incrementally. The internal pressure is used as a load parameter and is increased from zero to the maximum pressure $p_{\max}$. For n_{inc} load increments, the pressure at load increment $n+1$ is

$$p_{n+1} = \frac{n+1}{n_{\text{inc}}} p_{\max}.$$

At the beginning of each load increment, the displacement vector from the previous converged increment and the plastic strain history at the Gauss points are known. The displacement vector at the current pressure is then solved using the Newton-Raphson method.

In Newton iteration k, the linearized equilibrium equation is

$$\mathbf{K}_t^k \Delta \hat{\mathbf{u}}^k = -\hat{\mathbf{R}}^k.$$

The displacement vector is updated by

$$\hat{\mathbf{u}}^{k+1} = \hat{\mathbf{u}}^k + \Delta \hat{\mathbf{u}}^k.$$

During the Newton iterations of one load increment, the pressure p_{n+1} is kept constant. Only the displacement vector is updated until equilibrium is reached.

The Newton iteration is stopped when both convergence criteria are satisfied:

$$\left\| \hat{\mathbf{R}} \right\|_\infty < 0.005 \left\| \hat{\mathbf{F}}_{\text{int}} \right\|_\infty,$$

and

$$\left\| \Delta \hat{\mathbf{u}}^k \right\|_\infty < 0.005 \left\| \hat{\mathbf{u}} \right\|_\infty.$$

The plastic strain from the previous converged load increment is used as the old internal state variable during the Newton iterations. After convergence of the current load increment, the updated plastic strain is committed:

$$\boldsymbol{\varepsilon}_n^{pl} \leftarrow \boldsymbol{\varepsilon}_{n+1}^{pl}.$$

This committed plastic strain field is then used as the history variable for the next load increment. Therefore, the path-dependent elastic-plastic response is captured by transferring the Gauss-point plastic strain history from one load increment to the next.

7 Program Structure

The implementation is divided into separate Python files according to the numerical workflow. The program starts from the main file, reads the input data, creates the mesh, performs the finite element solution and finally generates the required plots.

The general program flow is

```
                main_elastic.py / main_plastic.py
                              ↓
                 run_elastic.py / run_plastic.py
                              ↓
          solver_routine_elastic.py / solver_routine_plastic.py
                              ↓
       assembly_routine_elastic.py / assembly_routine_plastic.py
                              ↓
        element_routine_elastic.py / element_routine_plastic.py
                              ↓
       material_routine_elastic.py / material_routine_plastic.py
```

The purpose of each file is summarized in Table 2.

Table 2: Overview of the implemented Python files.

File	Purpose
`input_data.py`	Defines material, geometry, loading and solver parameters
`mesh.py`	Generates radial mesh, connectivity and assignment matrix
`kinematics.py`	Computes shape functions, Jacobian, B-matrix and strain
`material_routine_elastic.py`	Computes elastic stress and elastic tangent matrix
`material_routine_plastic.py`	Performs return mapping and computes consistent tangent
`element_routine_elastic.py`	Computes elastic element force and stiffness
`element_routine_plastic.py`	Computes plastic element force, stiffness and state variables
`assembly_routine_elastic.py`	Assembles global elastic force vector and stiffness matrix
`assembly_routine_plastic.py`	Assembles global plastic system and Gauss-point quantities
`solver_routine_elastic.py`	Solves the elastic problem using Newton-Raphson method
`solver_routine_plastic.py`	Solves each plastic load increment using Newton-Raphson method
`run_elastic.py`	Runs elastic verification and elastic mesh convergence
`run_plastic.py`	Runs plastic simulation and convergence studies
`main_elastic.py`	Starts elastic analysis and creates elastic plots
`main_plastic.py`	Starts plastic analysis and creates plastic plots

For the elastic analysis, the solver is called once at the verification pressure p_{elastic}. For the plastic analysis, the pressure is applied incrementally from zero to $p_{\max}$. At every load increment, the Newton-Raphson solver calls the assembly routine, which loops over all elements. Each element routine evaluates the strain-displacement matrix, calls the material routine, computes the element internal force and tangent stiffness, and returns the Gauss-point quantities required for the plastic history and post-processing.

8 User Manual

The program was implemented in Python. The required software environment is:

- Python 3.11
- NumPy
- Matplotlib

All input parameters are defined in the file `input_data.py`. This includes the material parameters, geometry, maximum pressure, number of elements, number of load increments, convergence tolerances and mesh refinement factor.

To run the elastic verification, execute:

```
python main_elastic.py
```

This performs the elastic simulation at p_{elastic}, compares the FEM displacement with the analytical elastic solution, performs the elastic mesh convergence study and saves the elastic plots.

To run the elastic-plastic simulation, execute:

```
python main_plastic.py
```

This performs the full elastic-plastic simulation up to $p_{\max}$, evaluates the final displacement and stress distributions, performs the mesh and load-increment convergence studies, and saves the plastic plots.

The elastic output plots are saved in

```
elastic_plots/
```

and the plastic output plots are saved in

```
plastic_plots/
```

The main numerical outputs are printed in the terminal and include the final outer displacement $u_r(b)$, the final number of plastic Gauss points, Newton iteration information and convergence-study values.

9 Verification and Results

9.1 Elastic verification

The elastic verification was performed at

$$p_{\text{elastic}} = 37.5\,\text{MPa}.$$

This pressure is below the plastic initiation pressure

$$p_{\text{init}} = 75\,\text{MPa},$$

so the material response remains purely elastic. Therefore, the FEM solution can be compared with the analytical elastic solution for a thick-walled pipe,

$$u_r^{\text{elast}} = \frac{p}{E}\frac{a^2}{b^2 - a^2}\left[(1-\nu)r + (1+\nu)\frac{b^2}{r}\right].$$

For $n_{\text{elem}} = 20$, the computed FEM displacements at the inner and outer radii were

$$u_r(a) = 0.011\,062\,5\,\text{mm}, \qquad u_r(b) = 0.0075\,\text{mm}.$$

The corresponding analytical values were identical up to numerical precision. The relative infinity norm error was

$$\frac{\|\hat{\mathbf{u}}_{\text{FEM}} - \hat{\mathbf{u}}_{\text{exact}}\|_\infty}{\|\hat{\mathbf{u}}_{\text{exact}}\|_\infty} \approx 1.88 \times 10^{-15}.$$

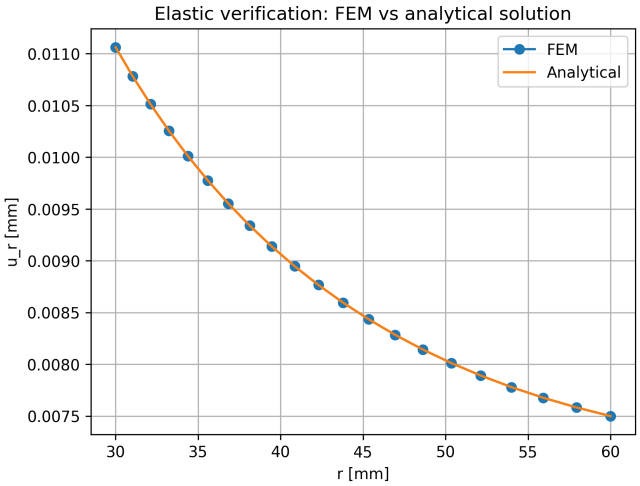


Figure 1: Comparison of the FEM displacement solution with the analytical elastic solution.

The FEM solution agrees with the analytical solution with machine-level accuracy. The mesh convergence study in Fig. 2 confirms that the error remains very small for all tested meshes. Since the errors are already of the order of 10^{-15}, the apparent variation in the convergence curve is dominated by floating-point roundoff rather than discretization error.

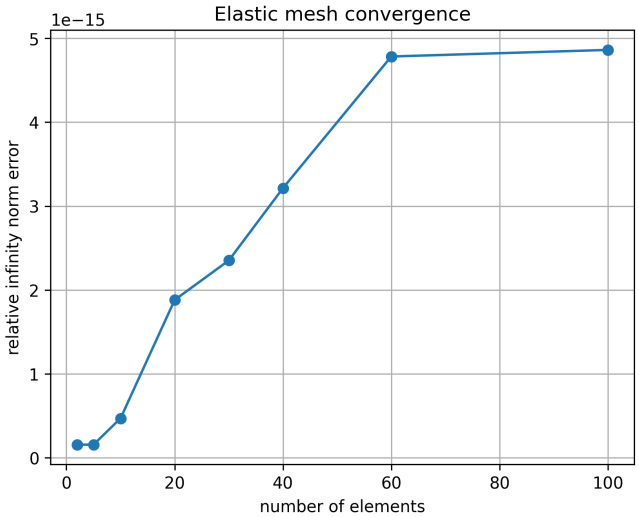


Figure 2: Elastic mesh convergence using the relative infinity norm error.

The implementation reports two Newton iterations for the elastic problem. However, the first Newton correction already reaches the linear solution. The second correction is negligible

and only occurs because both the residual criterion and the displacement-increment criterion are checked.

9.2 Plastic response at final loading

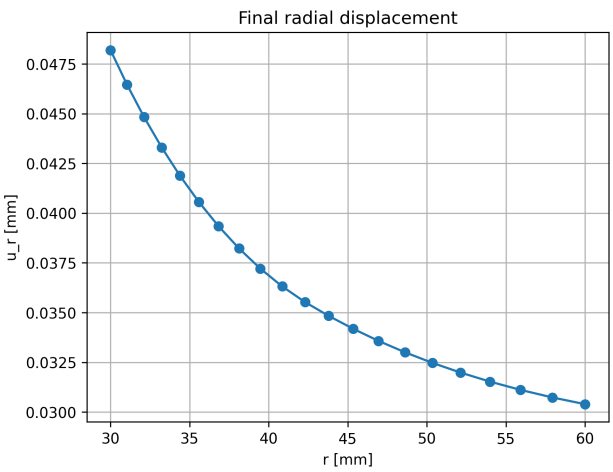
The full elastic-plastic simulation was performed by increasing the internal pressure gradually from zero to

$$p_{\max} = 120\,\mathrm{MPa}.$$

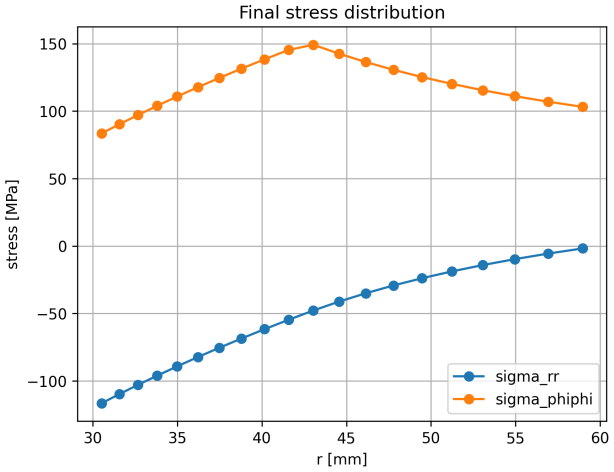
Since $p_{\max} > p_{\mathrm{init}}$, plastic deformation occurs during the loading process. For the reference simulation with $n_{\mathrm{elem}} = 20$ and $n_{\mathrm{inc}} = 50$, the final outer displacement was

$$u_r(b) \approx 0.030\,397\,\mathrm{mm}.$$

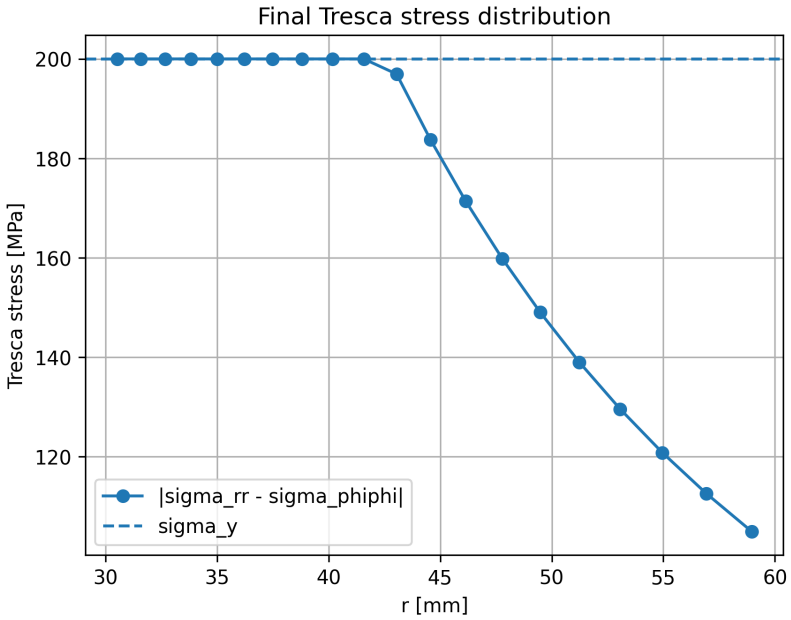
At $p_{\max}$, 10 of the 20 Gauss points were plastic, indicating that the plastic zone had propagated from the inner radius into the pipe wall while the outer region remained elastic.



(a) Radial displacement



(b) Stress distribution



(c) Tresca stress

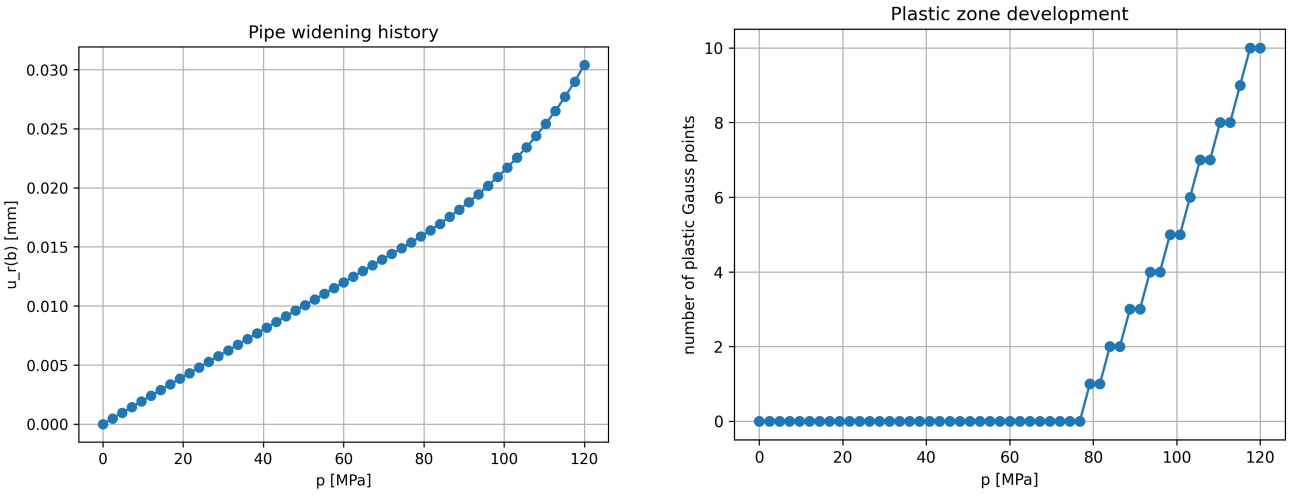
Figure 3: Final displacement and stress fields at $p_{\max} = 120\,\mathrm{MPa}$.

The radial displacement is largest at the inner radius and decreases towards the outer radius. This is expected because the pipe is loaded internally and the deformation is driven by the pressure acting at $r = a$.

The radial stress σ_{rr} is compressive and approaches zero at the traction-free outer boundary. The hoop stress $\sigma_{\varphi\varphi}$ is tensile and has its largest magnitude near the transition region between the plastic and elastic parts of the pipe wall. The Tresca stress distribution shows that the yield stress is reached in the plastic region, while the remaining outer part of the pipe stays elastic.

9.3 Pipe widening and plastic zone development

The widening of the pipe is measured by the radial displacement of the outer radius $u_r(b)$. Its evolution with increasing internal pressure is shown in Fig. 4.



(a) Pipe widening history

(b) Plastic zone development

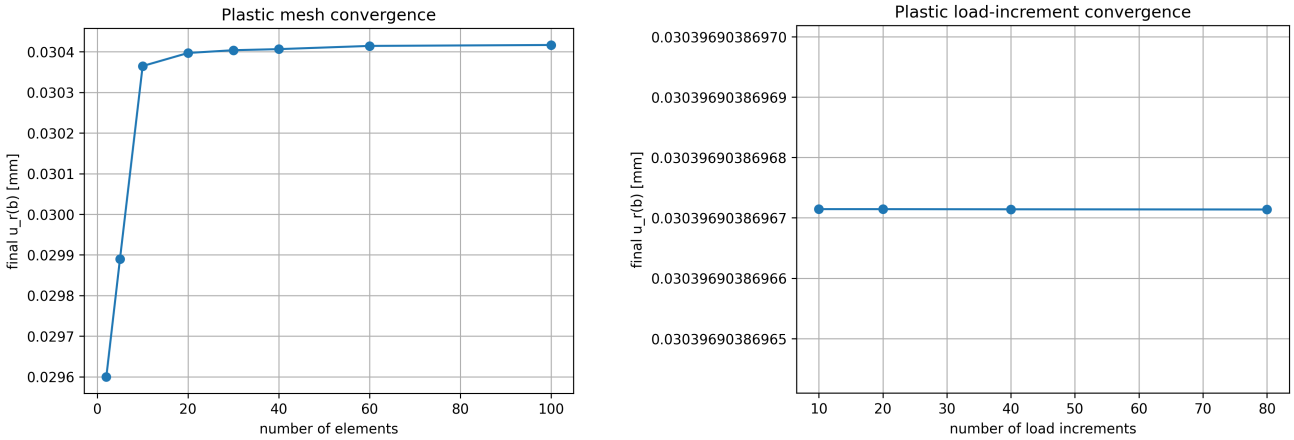
Figure 4: Load history of pipe widening and plastic zone evolution.

The pipe widening increases continuously with the applied pressure. In the lower pressure range, the response is almost linear because the whole pipe behaves elastically. Once the pressure exceeds the plastic initiation pressure, plastic deformation begins near the inner radius and the displacement response becomes nonlinear.

The plastic zone develops from the inner part of the pipe and grows outward as the pressure increases. The theoretical plastic initiation pressure is $p_{\text{init}} = 75\,\text{MPa}$. In the numerical simulation, the first detected plastic Gauss point can appear slightly above this value because the material response is evaluated at the Gauss point inside the first element and not exactly at the inner boundary $r = a$.

9.4 Plastic convergence studies

For the full plastic loading case, convergence studies were performed with respect to the number of elements and the number of load increments. The final outer displacement $u_r(b)$ was used as the convergence quantity.



(a) Mesh convergence

(b) Load-increment convergence

Figure 5: Plastic convergence studies using final outer displacement $u_r(b)$.

The mesh convergence study shows that the final pipe widening approaches a stable value as the number of elements is increased. The load-increment convergence study gives almost identical final values for $u_r(b)$ for all tested numbers of increments. This indicates that the chosen load discretization is sufficient for the present problem.

Overall, the elastic verification, plastic stress state, plastic zone development and convergence studies show that the implemented finite element solver gives physically consistent and numerically stable results.

10 Conclusion

A finite element solver for the axisymmetric elastic-plastic analysis of a thick-walled pipe under internal pressure was implemented in Python. The formulation uses linear finite elements in the radial direction, one Gauss point per element, a plane stress elastic law and an elastic ideal-plastic Tresca material model.

The elastic part of the implementation was verified using the analytical displacement solution for a thick-walled pipe. For the verification load $p_{\text{elastic}} = 37.5\,\text{MPa}$, the FEM solution matched the analytical solution with a relative infinity norm error of approximately 1.88×10^{-15}. This confirms the correctness of the spatial discretization, element routine, assembly procedure and boundary loading.

The full elastic-plastic simulation was performed up to $p_{\max} = 120\,\text{MPa}$. Since this pressure is larger than the plastic initiation pressure $p_{\text{init}} = 75\,\text{MPa}$, plastic deformation developed during loading. The final results show a physically consistent displacement field, stress distribution and Tresca stress distribution. The plastic zone started near the inner radius and grew outward with increasing pressure.

The convergence studies showed that the final pipe widening $u_r(b)$ converges with mesh refinement and is almost insensitive to the tested number of load increments. Therefore, the chosen discretization provides stable results for the considered variant.